

MH2100 Calculus III

Midterm II (Solutions)

AY 2019/2020, Semester 1

Quantitative Research Society @NTU

November 2019

Overview

- **Historical paper:** the original paper was held in November 2019 with **90 minutes** allowed.
- **Historical permitted material:** the November 2019 paper permitted a two-sided A4 page of handwritten notes. Consult current course materials for present assessment rules.
- **2026 scope note:** the current midterm scope ends at double integrals, including polar coordinates. Triple integrals, cylindrical and spherical coordinates, general substitutions, and line integrals are beyond that scope. They are retained and solved because they appear in this historical paper.

Instructions to candidates (historical)

1. DO NOT TURN OVER PAPER UNTIL INSTRUCTED.
2. This midterm paper contains THREE (3) questions.
3. Answer ALL questions. The marks for each question are indicated at the beginning of each question.
4. Candidates can write anywhere on this midterm paper.
5. One two-sided A4 piece of paper with handwritten notes is permitted.

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Question 1**(32 Marks)***(beyond 2026 midterm scope): Part (v) uses cylindrical coordinates.*

- (i) Use Lagrange Multipliers to find the maximum value of the function

$$f(x, y, z) = 14x + 8y + 12z$$

subject to the constraint $x^2 + y^2 + z^2 = 101$.

- (ii) Evaluate the iterated integral.

$$\int_{-1}^1 \int_0^1 ye^{xy} dx dy$$

- (iii) Find the volume of the solid bounded by the surface
- $z = 5 + (x - 4)^2 + 2y$
- and the planes
- $x = 0$
- ,
- $y = 0$
- ,
- $z = 0$
- ,
- $x = 3$
- ,
- $y = 4$
- .

- (iv) Evaluate the iterated integral by converting to polar coordinates.

$$\int_0^{\sqrt{\pi}} \int_0^{\sqrt{\pi-y^2}} \sin(x^2 + y^2) dx dy$$

- (v)
- (beyond 2026 midterm scope)*
- Use cylindrical coordinates to evaluate

$$\iiint_E y dV,$$

where E is the solid that lies between the cylinders $x^2 + y^2 = 3$ and $x^2 + y^2 = 7$, above the xy -plane and below the plane $z = x + 4$.**Solution****Method 1: Direct computation**

- (i) Let
- $g(x, y, z) = x^2 + y^2 + z^2$
- . The Lagrange equations are

$$(14, 8, 12) = \lambda(2x, 2y, 2z), \quad x^2 + y^2 + z^2 = 101.$$

Thus (x, y, z) is proportional to $(14, 8, 12)$; because

$$14^2 + 8^2 + 12^2 = 404 = 4 \cdot 101,$$

the point on the constraint in that direction is $(7, 4, 6)$. It gives

$$f(7, 4, 6) = 98 + 32 + 72 = 202.$$

Moreover, Cauchy–Schwarz gives, for every feasible point,

$$f(x, y, z) \leq \sqrt{14^2 + 8^2 + 12^2} \sqrt{x^2 + y^2 + z^2} = \sqrt{404} \sqrt{101} = 202,$$

with equality at $(7, 4, 6)$. The opposite point gives -202 , so

$\max f = 202, \quad \text{attained at } (7, 4, 6).$
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(ii) For $y \neq 0$, integrate first in x :

$$\int_0^1 y e^{xy} dx = [e^{xy}]_{x=0}^{x=1} = e^y - 1.$$

The resulting expression also gives the continuous value at $y = 0$. Hence

$$\int_{-1}^1 (e^y - 1) dy = [e^y - y]_{-1}^1 = e - e^{-1} - 2.$$

$$\boxed{\int_{-1}^1 \int_0^1 y e^{xy} dx dy = e - e^{-1} - 2.}$$

(iii) This is the volume below the graph and above $z = 0$ over the rectangle $0 \leq x \leq 3$, $0 \leq y \leq 4$. Therefore

$$V = \int_0^3 \int_0^4 (5 + (x-4)^2 + 2y) dy dx.$$

Separate the terms:

$$V = 4 \int_0^3 (5 + (x-4)^2) dx + 3 \int_0^4 2y dy.$$

Now

$$\int_0^3 (5 + (x-4)^2) dx = 15 + \left[\frac{(x-4)^3}{3} \right]_0^3 = 36,$$

so $V = 4(36) + 3(16) = 192$. Thus

$$\boxed{V = 192.}$$

(iv) The Cartesian region is the quarter-disk $x^2 + y^2 \leq \pi$ in the first quadrant. In polar coordinates it is

$$0 \leq r \leq \sqrt{\pi}, \quad 0 \leq \theta \leq \frac{\pi}{2}.$$

Hence

$$\int_0^{\pi/2} \int_0^{\sqrt{\pi}} \sin(r^2) r dr d\theta = \frac{\pi}{2} \left[-\frac{\cos(r^2)}{2} \right]_0^{\sqrt{\pi}} = \frac{\pi}{2}.$$

$$\boxed{\frac{\pi}{2}.$$

(v) *Beyond 2026 midterm scope.* Set $x = r \cos \theta$ and $y = r \sin \theta$. The solid has

$$\sqrt{3} \leq r \leq \sqrt{7}, \quad 0 \leq \theta \leq 2\pi, \quad 0 \leq z \leq r \cos \theta + 4.$$

The upper bound is positive because $4 - \sqrt{7} > 0$. Thus

$$\iiint_E y dV = \int_0^{2\pi} \int_{\sqrt{3}}^{\sqrt{7}} \int_0^{r \cos \theta + 4} r^2 \sin \theta dz dr d\theta.$$

After the z -integration, the integrand is

$$r^3 \sin \theta \cos \theta + 4r^2 \sin \theta.$$

Both terms integrate to zero over a full period in θ . Hence

$$\boxed{\iiint_E y dV = 0.}$$

Unofficial QRS marking scheme (32 marks)

- (i) **(8 marks)** 3 marks for the Lagrange system; 2 for the feasible point $(7, 4, 6)$; 1 for a compactness or Cauchy–Schwarz global-maximum justification; 1 for evaluating f ; 1 for identifying the maximum.
- (ii) **(6 marks)** 3 marks for the inner integral; 2 for the outer integral; 1 for the correct final value.
- (iii) **(6 marks)** 2 marks for the rectangular double-integral setup; 3 for correct evaluation; 1 for 192.
- (iv) **(6 marks)** 2 marks for the quarter-disk polar bounds; 2 for the Jacobian; 1 for the radial integral; 1 for $\pi/2$.
- (v) **(6 marks)** 2 marks for the cylindrical bounds; 2 for including $y = r \sin \theta$ and $dV = r dr d\theta dz$; 1 for the full-period cancellation; 1 for 0.

Question 2**(34 Marks)**

(beyond 2026 midterm scope): Part (i) uses spherical integration; part (iv) uses a general Jacobian substitution.

- (i) (beyond 2026 midterm scope) Use spherical coordinates to evaluate

$$\iiint_B \sqrt{x^2 + y^2 + z^2} \, dV,$$

where B is the ball $x^2 + y^2 + z^2 \leq 6$.

- (ii) Find the area of the region bounded by the curves $y = \sqrt{x+1}$ and $x = 3y - 1$.
 (iii) Evaluate the iterated integral

$$\int_0^{\sqrt{\pi/2}} \int_y^{\sqrt{\pi/2}} \cos(x^2) \, dx \, dy.$$

- (iv) (beyond 2026 midterm scope) Use the transformation $x = u/v$; $y = v$ to evaluate the integral

$$\iint_R xy^3 \, dA,$$

where R is the region in the first quadrant ($x \geq 0$ and $y \geq 0$) bounded by the lines $y = x$, $y = 3x$, and the hyperbolas $xy = 1$ and $xy = 4$.

- (v) Find the volume of the solid inside the cylinder $x^2 + y^2 = 16$ and the ellipsoid $2x^2 + 2y^2 + z^2 = 32$.

Solution**Method 1: Direct computation**

- (i) Beyond 2026 midterm scope. In spherical coordinates the integrand is ρ , the volume factor is $\rho^2 \sin \phi$, and the ball has $0 \leq \rho \leq \sqrt{6}$. Therefore

$$\iiint_B \sqrt{x^2 + y^2 + z^2} \, dV = \int_0^{2\pi} \int_0^\pi \int_0^{\sqrt{6}} \rho^3 \sin \phi \, d\rho \, d\phi \, d\theta = 4\pi \left[\frac{\rho^4}{4} \right]_0^{\sqrt{6}}.$$

Thus

$$\boxed{36\pi}.$$

- (ii) Write the first curve as $x = y^2 - 1$. The intersections satisfy

$$y^2 - 1 = 3y - 1, \quad y(y - 3) = 0,$$

so $0 \leq y \leq 3$. On this interval, $y^2 - 1 \leq x \leq 3y - 1$. Hence the area is

$$A = \int_0^3 [(3y - 1) - (y^2 - 1)] \, dy = \int_0^3 (3y - y^2) \, dy = \left[\frac{3y^2}{2} - \frac{y^3}{3} \right]_0^3.$$

Therefore

$$\boxed{A = \frac{9}{2}}.$$

- (iii) The region is $0 \leq y \leq x \leq \sqrt{\pi/2}$. Reverse the order:

$$\int_0^{\sqrt{\pi/2}} \int_0^x \cos(x^2) dy dx = \int_0^{\sqrt{\pi/2}} x \cos(x^2) dx = \left[\frac{1}{2} \sin(x^2) \right]_0^{\sqrt{\pi/2}}.$$

Consequently,

$$\boxed{\frac{1}{2}}.$$

- (iv) *Beyond 2026 midterm scope.* The inverse transformation is

$$u = xy, \quad v = y,$$

and

$$\left| \frac{\partial(x, y)}{\partial(u, v)} \right| = \left| \begin{vmatrix} 1/v & -u/v^2 \\ 0 & 1 \end{vmatrix} \right| = \frac{1}{v}.$$

The hyperbolas give $1 \leq u \leq 4$. Since $y/x = v^2/u$, the two lines give

$$\sqrt{u} \leq v \leq \sqrt{3u}.$$

Moreover, $xy^3 = (u/v)v^3 = uv^2$. Thus

$$\iint_R xy^3 dA = \int_1^4 \int_{\sqrt{u}}^{\sqrt{3u}} uv dv du = \int_1^4 u^2 du = \left[\frac{u^3}{3} \right]_1^4.$$

Hence

$$\boxed{21}.$$

- (v) Over the disk $x^2 + y^2 \leq 16$, the ellipsoid lies between the graphs $z = \pm \sqrt{32 - 2(x^2 + y^2)}$. Its radial extent is $0 \leq r \leq 4$, exactly the cylinder's interior. The volume is the polar double integral of the vertical height:

$$V = \int_0^{2\pi} \int_0^4 2\sqrt{32 - 2r^2} r dr d\theta.$$

Thus

$$V = 4\pi \int_0^4 r\sqrt{32 - 2r^2} dr = 4\pi \left[-\frac{(32 - 2r^2)^{3/2}}{6} \right]_0^4.$$

Therefore

$$\boxed{V = \frac{256\sqrt{2}\pi}{3}}.$$

Unofficial QRS marking scheme (34 marks)

- (i) **(7 marks)** 2 marks for spherical bounds; 2 for the integrand and Jacobian; 2 for integration; 1 for 36π .
- (ii) **(6 marks)** 2 marks for intersections; 2 for correct horizontal bounds; 1 for integration; 1 for $9/2$.
- (iii) **(6 marks)** 3 marks for reversed bounds; 2 for the substitution $u = x^2$; 1 for $1/2$.
- (iv) **(7 marks)** 2 marks for the transformed region; 2 for the Jacobian; 2 for the transformed integral; 1 for 21.
- (v) **(8 marks)** 2 marks for the vertical height and polar double-integral setup; 2 for identifying the common radial endpoint $r = 4$; 3 for integration; 1 for $256\sqrt{2}\pi/3$.

Question 3**(34 Marks)***(beyond 2026 midterm scope): All parts use scalar or vector line integrals.*

- (i)
- (beyond 2026 midterm scope)*
- Evaluate the line integral

$$\int_C xy \, ds,$$

where C is the line segment from $(-2, -1)$ to $(4, 5)$.

- (ii)
- (beyond 2026 midterm scope)*
- Find the work done by the force field
- $\mathbf{F}(x, y) = y\mathbf{i} - x\mathbf{j}$
- on a particle that moves along the curve
- $x = 2y^3 - 6y^2 + 1$
- from
- $(-3, 1)$
- to
- $(1, 3)$
- .

- (iii)
- (beyond 2026 midterm scope)*
- Evaluate the line integral

$$\int_C x^2 y \cos z \, ds,$$

where C has parametrisation $x = \cos t$, $y = \sin t$ and $z = t$, with $0 \leq t \leq \pi$.

- (iv)
- (beyond 2026 midterm scope)*
- Let
- $\mathbf{F}(x, y) = y\mathbf{i} + (x + 10y)\mathbf{j}$
- and let
- C
- be the upper semicircle that starts at
- $(1, 2)$
- and ends at
- $(5, 2)$
- .

(1) Find a function f such that $\mathbf{F} = \nabla f$.(2) Hence, or otherwise, evaluate the line integral $\int_C \mathbf{F} \cdot d\mathbf{r}$.**Solution****Method 1: Parametrise directly; use a potential where available**

- (i)
- Beyond 2026 midterm scope.*
- Parametrise the segment by

$$\mathbf{r}(t) = (-2 + 6t, -1 + 6t), \quad 0 \leq t \leq 1.$$

Then $\|\mathbf{r}'(t)\| = 6\sqrt{2}$, so

$$\int_C xy \, ds = 6\sqrt{2} \int_0^1 (-2 + 6t)(-1 + 6t) \, dt = 6\sqrt{2} \int_0^1 (2 - 18t + 36t^2) \, dt.$$

Therefore

$$\boxed{\int_C xy \, ds = 30\sqrt{2}.}$$

- (ii)
- Beyond 2026 midterm scope.*
- Use
- y
- as the parameter, with
- $1 \leq y \leq 3$
- and
- $x = 2y^3 - 6y^2 + 1$
- . Since

$$\frac{dx}{dy} = 6y^2 - 12y,$$

the work is

$$W = \int_1^3 (y \, dx - x \, dy) = \int_1^3 [y(6y^2 - 12y) - (2y^3 - 6y^2 + 1)] \, dy.$$

Thus

$$W = \int_1^3 (4y^3 - 6y^2 - 1) dy = [y^4 - 2y^3 - y]_1^3.$$

Hence

$$\boxed{W = 26.}$$

(iii) *Beyond 2026 midterm scope.* With

$$\mathbf{r}(t) = (\cos t, \sin t, t),$$

we have $\|\mathbf{r}'(t)\| = \sqrt{2}$. Consequently,

$$\int_C x^2 y \cos z \, ds = \sqrt{2} \int_0^\pi \cos^3 t \sin t \, dt = -\frac{\sqrt{2}}{4} [\cos^4 t]_0^\pi.$$

The endpoint values are equal, so

$$\boxed{\int_C x^2 y \cos z \, ds = 0.}$$

(iv) *Beyond 2026 midterm scope.*

(1) From $f_x = y$, integrate in x to obtain $f(x, y) = xy + h(y)$. Comparing

$$f_y = x + h'(y)$$

with $x + 10y$ gives $h'(y) = 10y$, hence one suitable potential is

$$\boxed{f(x, y) = xy + 5y^2.}$$

(2) The endpoints are $(1, 2)$ and $(5, 2)$. The fundamental theorem for line integrals gives

$$\int_C \mathbf{F} \cdot d\mathbf{r} = f(5, 2) - f(1, 2) = (10 + 20) - (2 + 20).$$

Thus

$$\boxed{\int_C \mathbf{F} \cdot d\mathbf{r} = 8.}$$

Unofficial QRS marking scheme (34 marks)

- (i) **(8 marks)** 3 marks for a valid parametrisation; 2 for $ds = 6\sqrt{2} dt$; 2 for integration; 1 for $30\sqrt{2}$.
- (ii) **(8 marks)** 2 marks for parametrising with y ; 3 for $\mathbf{F} \cdot d\mathbf{r}$; 2 for integration; 1 for 26.
- (iii) **(8 marks)** 2 marks for the parametrisation; 2 for $ds = \sqrt{2} dt$; 3 for the scalar line integral; 1 for 0.
- (iv) **(10 marks)**
 - (1) 5 marks: 3 marks for integrating f_x ; 2 marks for determining $h(y)$.
 - (2) 5 marks: 2 marks for the endpoints; 2 marks for applying the potential-difference theorem; 1 mark for 8.